

Association of serum LDL cholesterol level with periodontitis among patients visiting a tertiary-care hospital.

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INTRODUCTION: High low-density lipoproteins (LDL) cholesterol is one of the major risk factors for cardiovascular disease. In recent years, some evidence has been presented that periodontitis, an infectious inflammatory condition of the periodontium, is associated with an increased risk of cardiovascular disease. To further elucidate this association, we have studied the levels of LDL cholesterol, a known risk marker for cardiovascular disease, in a periodontally-diseased group. METHODS: The levels of serum LDL cholesterol in 47 subjects with mild to severe (clinical attachment loss equal to or greater than 1 mm) chronic generalized (at least 30% of teeth affected) periodontitis with the mean age of 42.21 ± 1.46 years were measured and compared with those obtained from 42 age (39.83 ± 0.94) and sex matched controls. Both groups were free from systemic illnesses. RESULTS: The mean serum LDL cholesterol in periodontitis patients was found to be significantly higher ($P < 0.001$) as compared to that of the controls. The mean clinical attachment loss was positively correlated with serum LDL cholesterol ($P < 0.01$) and gingival index ($P < 0.05$). The frequency of persons with pathologic values of LDL cholesterol was significantly higher in periodontitis patients compared with that of the controls. CONCLUSIONS: These results showed that high serum LDL cholesterol may be associated with periodontitis in healthy people. However, it is unclear whether periodontitis causes an increase in the levels of serum LDL or an increased LDL is a risk factor for both periodontitis and cardiovascular disease.